Project Phase 5: Architecting for Scale

Due Date: End of Week 8 **Points:** 100

Scenario

Incredible news! The Riverside Cafe is going to be featured on a famous TV food show. This is a massive opportunity, but it also presents a huge technical challenge. A massive, temporary spike in website traffic is expected, which will overwhelm the single application server.

Your task is to re-architect the application tier to be fully scalable and highly available, ensuring the website remains fast and responsive throughout the traffic surge.

Project Goals

1. **Ensure High Availability.** Deploy your architecture across at least two Availability Zones to withstand the failure of a single data center.
2. **Distribute the Load.** Implement a load balancer to act as the single entry point for all web traffic, distributing requests evenly.
3. **Implement Elasticity.** Create an auto-scaling mechanism that automatically launches and terminates application servers in response to demand.

Technical Requirements

- Your VPC must be configured to support a multi-AZ deployment (subnets in at least two AZs).
- Your application servers must be part of an Auto Scaling group managed by a launch template.
- The Application Load Balancer should be placed in the public subnets, forwarding traffic to the application servers in the private subnets.

Submission Instructions

You will submit a single PDF document to Moodle. This document must contain all parts of the deliverable outlined below.

Deliverable Breakdown

- **The Scalability & Availability Plan:** What are the two primary risks of The Riverside Cafe's current single-server architecture in this scenario? Describe the three core cloud services/features you will use to solve this problem (Load Balancing, Auto Scaling, Multi-AZ) and explain how each contributes to the solution.
- **The Resilient Architecture:** Provide a new, detailed architecture diagram showing your scalable design. It must illustrate the VPC spanning two AZs, the Application Load Balancer, the Auto Scaling group managing instances across both AZs, and the user traffic flow.

- **Launch Template Configuration:** Explain the purpose of a launch template and why it is critical for auto-scaling. Provide a screenshot of your launch template's configuration summary.
- **Auto Scaling Policy Design:** Detail and justify your choices for the Auto Scaling group's Desired, Minimum, and Maximum capacity. What metric did you choose to trigger scaling events (e.g., Average CPU Utilization)? Justify your choice of metric and target value.
- **Verification and Testing Plan:** Provide the public DNS name of your load balancer. Describe the process you would use to simulate load and test your auto-scaling setup. Provide screenshots showing the "scale-out" event: one showing the metric alarm was triggered, and another showing new instances launched by the Auto Scaling group.

Grading Rubric

Criteria	Description	Points
Analysis & Plan	Risks are correctly identified and the proposed solution using the three core services is well-explained.	20
Architecture Diagram	Diagram is detailed, accurate, and clearly shows the multi-AZ, load-balanced, auto-scaled architecture.	25
Launch Template	Purpose of the launch template is understood and configuration is shown.	10
Auto Scaling Policy	Scaling parameters and policy are well-justified for the scenario.	20
Testing & Verification	A clear testing plan is described, and screenshot evidence proves the system scaled as designed.	25
Total		**100**